E-Waste and EPR Compliance Assessment of India's Leading Electronics Manufacturers

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Abstract

As electronic consumption grows exponentially across the Indian subcontinent, regulatory bodies such as the Central Pollution Control Board (CPCB) have tightened enforcement under the E-Waste (Management) Rules, 2022. While these laws outline strict legal obligations on paper, actual operational compliance varies significantly depending on a company's position in the hardware supply chain. For contract Electronic Manufacturing Services (EMS) providers, the regulatory burden differs from that of direct brand-owner Original Equipment Manufacturers (OEMs). To understand the practical execution of India's legislative framework, this section assesses the compliance disclosures of five leading electronics manufacturers currently operating in the domestic market.

Keywords: E-waste, Electronic Manufacturing Services, Original Equipment Manufacturers, Impacts, Recovery, Recycling, Technology.

1. Kaynes Technology India Limited

Kaynes Technology is a major player in design-led electronics manufacturing services in India. As a publicly traded company, it is mandated to submit detailed Business Responsibility and Sustainability Reports (BRSR) to SEBI, providing highly transparent environmental data.

Followed Rules:

- **E-Waste (Management) Rules, 2016 / 2022 (Disposal):** Kaynes officially reports that all generated internal e-waste (amounting to 4.96 Metric Tonnes in FY 2022-23, up from 3.86 MT in FY 2021-22) is successfully collected, logged, and transferred to authorized state-registered recycling and dismantling agencies under EPR guidelines.
- **Restriction of Hazardous Substances (RoHS) Directive:** Ensures that 100% of its printed circuit board assemblies are strictly RoHS and REACH compliant, acting as a responsible component supplier to global automotive and industrial OEMs.
- **Air and Water Pollution Prevention Acts:** Complies fully with zero pending statutory non-compliance cases filed by State Pollution Control Boards. Regularly monitors factory emissions (such as NOx at 9.56 ppmv and SOx at 4.09 ppmv) through independent labs.
- **Factories Act, 1948 and OHSAS:** Maintains fully certified environmental health and safety (EHS) policies. Implements Hazard Identification and Risk Assessment (HIRA) and safety audits audited and certified by Bureau Veritas and Intertek.

Not Followed Rules (or Not Applicable):

- **E-Waste (Management) Rules (Direct Consumer EPR):** Not Applicable. Because Kaynes operates on a pure business-to-business (B2B) EMS contract manufacturing model, direct retail-level consumer take-back and collection targets are not applicable to their activities (the legal burden lies on brand owners/OEMs).
- **Rights of Persons with Disabilities Act, 2016:** Officially acknowledges in its SEBI BRSR filings that its manufacturing facilities and offices are not yet fully physically accessible to differently-abled employees, and the firm does not have a registered Equal Opportunity Policy.
- **Circular Recovery and Sustainable Sourcing Gaps:** Reports 0 MT of internally recycled or re-used raw inputs, showing zero in-house circular reclamation. Lacks formalized procedures for sustainable sourcing and human rights clauses in standard supply chain vendor contracts.

2. Sahasra Electronics Private Limited

Sahasra Electronics provides comprehensive electronic manufacturing services across industrial and consumer sectors. To fulfill its e-waste obligations under national rules, the company has structured an outsource-led compliance framework.

Followed Rules:

- **E-Waste (Management) Rules, 2016 & 2018 (EPR Plan):** Fully implements a structured EPR Plan authorized by the state pollution boards. It contractually delegates its physical waste management, dismantling, and recycling obligations to M/s E-Waste Recyclers India, a Haryana-certified registered recycler.
- **Reverse Logistics and Customer Hotlines:** Maintains a nationwide reverse logistics agreement with M/s Om Logistics Limited to safely secure the collection supply chain, and operates a public toll-free helpline (1800-102-5679) with online drop-off center maps.

Not Followed Rules (or Not Applicable):

- **Direct Proprietary Infrastructure Rules:** The company does not establish or manage any native, company-owned physical collection centers or recycling facilities, completely outsourcing the entire collection, transport, and tracing infrastructure to its partners.

3. Syrma SGS Technology Limited

Syrma SGS is a major Indian contract hardware manufacturer. It maintains a dual strategy of integrating certified recyclers while establishing structured collection points across state borders.

Followed Rules:

- **E-Waste (Management) Rules, 2016 / 2022:** Maintains standard regulatory alignment by partnering with M/s Global Waste Solution, an authorized recycler and dismantler certified by the State Pollution Control Board.
- **EPR State-Wise Network Coverage:** Operates an organized network of physical e-waste collection points in major industrial regions (including Tamil Nadu, Karnataka, Maharashtra, Delhi, and West Bengal), supported by a toll-free hotline (1800-270-7666) and clear 'Do's and Don'ts' guidelines in product manuals.

Not Followed Rules (or Not Applicable):

- **Localized Territorial Collection Gaps:** Does not maintain physical collection points in every single state. Instead, they route waste from minor uncovered states to centers located in neighboring states, creating localized transport risks and compliance lag.

4. Dixon Technologies (India) Limited

As India's largest domestic EMS provider, Dixon Technologies handles high volumes of industrial components. Its waste management focuses heavily on factory-floor safety and specialized chemical handling.

Followed Rules:

- **Factories Act, 1948 & Hazardous Waste Rules:** Maintains strict factory compliance. They partner with state-authorized treatment, storage, and disposal facilities (TSDF) such as Steam Oil & General Industries to safely incinerate and process hazardous industrial oil waste.
- **Internal Waste Monitoring:** Successfully audits and logs e-waste generated directly within its assembly facilities, preventing toxic leakage on-site.

Not Followed Rules (or Not Applicable):

- **Direct Post-Consumer EPR Collection:** Not Applicable. Operating under a pure B2B EMS contract model, Dixon has no direct consumer-facing buyback or retail collection systems, as post-consumer recycling obligations are legally borne by their OEM clients.

5. Tata Electronics Private Limited

Tata Electronics operates cutting-edge semiconductor design and EMS facilities. In line with the Tata Group's long-term sustainability goals, the company pursues aggressive 'Beyond Compliance' environmental policies.

Followed Rules:

- **Zero Liquid Discharge (ZLD) Model:** Exceeds basic regulatory baselines by operating advanced effluent treatment systems that recycle more than 6,50,000 KL of wastewater annually at its manufacturing hubs.
- **Zero Waste to Landfill:** Diverted 396 MT of hazardous manufacturing waste from landfills in FY 2023-24 through eco-friendly co-processing rather than burial or simple incineration.
- **Green Building Laws:** The flagship Hosur EMS plant is officially certified as a Platinum-rated Green Factory by the Indian Green Building Council (IGBC).

Not Followed Rules (or Not Applicable):

- **Public Consumer-Facing Infrastructure:** The company does not maintain public consumer-facing trade-in programs or retail-level e-waste collection bins, focusing its circular economy goals strictly on factory-floor and water resource sustainability.

6. Comparative Compliance Analysis

To provide a clear, aggregated summary of regulatory adherence across these top manufacturers, the matrix below maps their business models, primary followed legislation, and notable compliance gaps.

Table No. 3: Comparative E-Waste & Regulatory Compliance Matrix of Top Manufacturers

Source: Compiled from SEBI BRSR reports and official corporate e-waste filings.

Manufacturer	Business Model	Key Laws Followed	Disclosed Gaps / Not Applicable Rules
Kaynes Tech	EMS (B2B)	E-Waste 2016/2022 (Internal Disposal), Factories Act, OHSAS, Air & Water Acts (zero pending cases), RoHS Compliance (100% of PCBA).	Direct consumer collection (Not Applicable). Lacks physical accessibility under Rights of Persons with Disabilities Act. 0 MT in-house recycled.
Sahasra Elec.	EMS / Memory	E-Waste Rules 2016/2018 (Formal registered EPR Plan). Reverse logistics via Om Logistics. Partnerships with Haryana-registered M/s E-Waste Recyclers India.	Lacks proprietary collection centers or recycling facilities; fully outsources physical retrieval and recycling infrastructure.
Syrma SGS	EMS (B2B/B2C)	E-Waste Rules 2016/2022. State-wise collection network in major hubs (Tamil Nadu, Delhi, West Bengal). Partners with certified M/s Global Waste Solution.	Lacks native collection points in every state (routes minor states to neighbors).
Dixon Tech	EMS (B2B)	Factories Act, Hazardous Waste Rules. Disposal of production oils via state-authorized Steam Oil & General Industries. Internal factory e-waste logs.	Direct consumer-facing EPR (Not Applicable). Lacks retail buyback networks or consumer e-waste collection bins.
Tata Elec.	EMS / Fab	Zero Liquid Discharge (6.5 Lakh KL wastewater recycled). Zero Waste to Landfill (396 MT hazardous waste diverted). Platinum-rated Green Factory (IGBC).	Public retail recycling infrastructure or public consumer buyback programs.

10. Findings, Conclusion and Recommendations

Undoubtedly, India possesses an extensive set of environmental rules on paper. However, successful enforcement remains the major structural bottleneck. Because a significant proportion of electronic waste still flows through unorganized, unregulated local scrap dealers (Kabbadi networks), there is a critical gap in material recovery and public health safety. The following strategic actions are recommended to bridge these gaps and support a functional circular economy:

- **Integrate Organized and Unorganized Sectors:** Develop practical hybrid models where informal collectors are compensated to channel waste to authorized, capital-intensive dismantling centers rather than burning materials themselves.
- **Enforce Stringent Import Controls:** Empower the Central Board of Excise and Customs with the necessary technology and expert human resources to distinguish between imported second-hand products meant for refurbishment and illegal, toxic e-waste shipments.
- **Provide Financial and Fiscal Incentives:** Offer direct subsidies, tax breaks, or a dedicated environmental processing fee (funded by 2-5% of original device production costs) to registered, certified recycling firms based on audited volumes.
- **Deploy Digital Traceability:** Mandate that critical electronic sub-components, such as circuit boards, compressor units, and memory chips, are labeled with radio-frequency identification (RFID) tags to track downstream waste streams.
- **Implement Local Infrastructure and Awareness:** Upload comprehensive, transparent public directories of certified recyclers and collectors on all State Pollution Control Board websites, establish localized, separate sorting bins, and launch national deposit-refund systems for consumer purchases.

In conclusion, addressing the electronic waste crisis requires directing the same level of passion and creativity currently invested in designing new gadgets towards the responsible management of old ones. By aligning consumer behavior, manufacturer design, and regulatory monitoring with sustainable recycling technologies, India can turn a hazardous regulatory challenge into a thriving circular economy.

References

Biswas, A., & Ghanshyam Singh, S. (2020). E-waste Management in India: Challenges and Agenda. Centre for Science and Environment, New Delhi.

Central Pollution Control Board (CPCB). (2020). Report on e-waste management in India. Submitted to the National Green Tribunal in Original Application No. 1001 of 2019 (Shailesh Singh Vs State of Uttar Pradesh & Others).

Puckett, J., et al. (2002). Exporting Harm: The High-Tech Trashing of Asia. The Basel Action Network (BAN) and Silicon Valley Toxics Coalition (SVTC).

Duan, et al. (2018). Global warming potential and carbon dioxide equivalents of refrigerant leakage from consumer electronics. Journal of Environmental Management.

Jomova, K., Jenisova, Z., Feszterova, M., Baros, S., Liska, J., Hudecova, D., & Valko, M. (2011). Arsenic: toxicity, oxidative stress and human disease. Journal of Applied Toxicology, 31(2), 95–107.

Sinha-Khetriwal, D. (2005). The management of electronic waste: a comparative study on India and Switzerland. Environmental Impact Assessment Review, 25(5), 492-504.